AgriSmart: Low-Cost Sensor-Driven Crop Recommendation Using Temporal Consensus over a Relational Farm Database

Aarav Gupta
School of Computer Science and Engineering
Vellore Institute of Technology
Vellore, India
Reg. No. 25BAI0131

Chirag Saxena
School of Computer Science and Engineering
Vellore Institute of Technology
Vellore, India
Reg. No. 25BCE0223

Suyash Agarwal
School of Computer Science and Engineering
Vellore Institute of Technology
Vellore, India
Reg. No. 25BBS0088

***Abstract*—Crop recommenders in the literature usually classify a single soil sample that includes nitrogen, phosphorus and potassium (N-P-K) values, which a smallholder cannot measure cheaply. This paper presents AgriSmart, a crop recommendation system built around a normalised MySQL 8 database that is fed by an Arduino Uno node with only a soil-moisture sensor, an air temperature/humidity sensor and a soil-temperature probe; soil pH is optional. The database stores every reading, cultivation cycle and fertiliser treatment, and the recommendation is computed from this history instead of one reading. We propose (i) a quality-weighted 7-day temporal consensus (QW-TC) of readings, (ii) an automatic switch to a sensor-only model when pH is missing, (iii) isotonic calibration with Isolation-Forest abstention, (iv) a Virtual Nitrogen Balance Index estimated from crop and fertiliser history and implemented as an SQL function, and (v) an explainable per-crop fit card. The models are trained on a public 22-crop dataset in which soil moisture, soil temperature and soil type are derived with documented formulas. The deployed, calibrated soft-voting ensemble reaches 0.902 hold-out accuracy with pH, and a Random Forest reaches 0.814 without pH. Under 0–30 % simulated sensor faults, QW-TC keeps top-1 accuracy at 0.857–0.902, against 0.539–0.614 when only the latest reading is classified.**

Keywords—crop recommendation, relational database, Internet of Things, soil sensors, temporal consensus, model calibration, precision agriculture

I. Introduction

Indian farming cooperatives still keep land, soil-test, sowing and fertiliser records in disconnected registers and spreadsheets, so a plot's history cannot be linked to the yield it finally produced. Surveys of agricultural IoT name data heterogeneity and the absence of a unified schema as the main barrier to adoption [1], and note that farm data stays in silos [2].

Published crop recommenders add a further problem: they classify one soil sample that includes N, P and K [3]. In the component list of our first design, the RS485 N-P-K/pH probe alone cost INR 2,500 of an approximately INR 3,900 node. A single reading from a cheap sensor is also noisy, swings with the day/night cycle and sometimes faults; a disconnected DS18B20, for example, returns −127 °C [4]. Existing recommenders always return an answer, even for inputs unlike their training data, and they ignore what was grown on the plot before.

AgriSmart addresses these problems by making the relational database the centre of the system. A low-cost node without an N-P-K sensor writes readings into MySQL, and every recommendation is computed from the plot's stored history. The contributions are:

- **QW-TC:** a quality-weighted, day-blocked 7-day consensus of readings instead of a single snapshot;
- **adaptive model selection:** a separately trained sensor-only model when no valid pH exists;
- **calibrated, abstention-aware output:** isotonic calibration, an Isolation-Forest out-of-distribution check and a confidence threshold;
- **Virtual Nitrogen Balance Index (VNBI):** soil nitrogen estimated from crop and fertiliser history, implemented identically in SQL and Python, with season, rotation and nitrogen re-ranking;
- **explainable fit card:** each crop's learned P5–P95 range per measurement, stored in the database and shown with every recommendation.

Section II reviews related work, Section III describes the system and database, Section IV the dataset and models, Section V the proposed methods and Section VI the experiments. Sections VII and VIII discuss limitations and conclude.

II. Related Work

A. IoT Sensing Architectures

Ayaz *et al.* [5] surveyed sensors across the crop life-cycle and found soil moisture, pH and N-P-K to have the highest decision value per rupee, but did not discuss data storage. Elijah *et al.* [1] proposed a four-layer architecture whose analytics layer assumes the cloud, which is costly for cooperatives. Tzounis *et al.* [6] showed that a low-power microcontroller with duty-cycled sampling is sufficient for slowly changing soil parameters, and Khanna and Kaur [7] reported that the cost of a soil-sensing node has fallen to a level where cooperative adoption is feasible. Farooq *et al.* [8] recommended validating readings at the edge before they are stored. LoRaFarM [9] is low-cost and modular but stores data as a simple time-series log, and Villa-Henriksen *et al.* [10] found interoperability and record-keeping, not sensing accuracy, to be the main blockers.

B. Farm Data Management

Kamilaris *et al.* [11] found that most agricultural big-data projects lack a normalised, queryable core, and Bhat and Huang [12] list data integration as the foremost open

challenge. Sharma *et al.* [13] identify traceability of inputs to outputs as a precondition for supply-chain analytics. Zhang *et al.* [14] showed that within-field variability justifies plot-level management, and Dhanaraju *et al.* [15] reported that sensor-guided input dosing reduces fertiliser overuse.

C. Machine Learning in Agriculture

Liakos *et al.* [16] found soil management to be the least explored branch of agricultural machine learning. Deep learning outperforms classical methods but needs large labelled datasets that a cooperative does not have [17]. A systematic review of yield prediction [18] found that farm-level historical logs are rarely used as features. Random Forests gave markedly lower error than linear regression for yield [19]; a small set of soil and climate features explains most yield variance [20]; and a deep reinforcement learning model showed that past seasons carry predictive signal, at the cost of interpretability [21].

D. Irrigation, Crop Recommendation and Rotation

Sensor networks for irrigation [22]–[24] use measured soil moisture for threshold or closed-loop control, and Muangprathub *et al.* [25] derived moisture bands from stored history. Pudumalar *et al.* [3] recommended crops with a voting ensemble that outperformed single classifiers, but from one-off soil samples with no history. Fenz [26] planned crop rotations from multi-year field records, but the planner works offline and is not enforced at data entry.

E. Research Gaps

From this review we identified five gaps: (1) sensed soil data is not linked in a normalised schema to the plot, crop cycle and yield; (2) crop recommendation ignores soil history; (3) crop rotation is planned but never enforced; (4) fertiliser and pesticide use is not traceable to output; and (5) results are reported as model accuracy without simple metrics that a clerk can verify by hand. AgriSmart closes gaps 1, 3, 4 and 5 with schema objects (tables, a trigger, views and a stored procedure) and gap 2 with QW-TC.

III. System Design

Fig. 1 shows the architecture. Readings travel from the sensor node over USB to a Python serial bridge, which inserts them into MySQL. A Flask website reads the database and calls the recommendation engine.

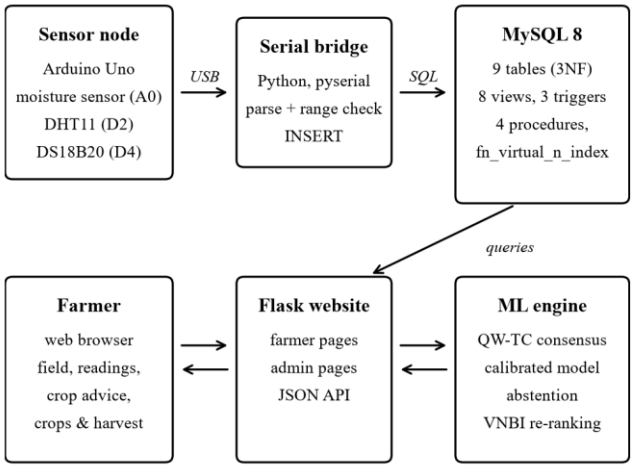


Fig. 1. AgriSmart architecture.

A. Sensor Node

The node is an Arduino Uno with a capacitive soil-moisture sensor on A0, a DHT11 air temperature and humidity sensor on D2 [27] and a DS18B20 waterproof soil-temperature probe with a 4.7 kΩ pull-up on D4 [4]. A PH-4502C pH module on A1 is optional; otherwise pH is typed in from the farmer's Soil Health Card. The sketch prints a header line and then one CSV line per sample (node_id, moisture, soil temperature, air temperature, humidity, pH), writing "nan" for a missing value. Moisture uses a two-point calibration between a dry and a saturated reading (default saturated water content 45 %). Dropping the N-P-K probe and its MAX485 interface from the Review-1 component list leaves parts listed at about INR 1,380.

B. Serial Bridge

The bridge reads the column order from the sketch's header line and accepts 6-field or 4-field frames. It maps "nan" to NULL, sets values outside the physical range (moisture 0–60 %, soil temperature −5–60 °C, air temperature 0–50 °C, humidity 5–100 %, pH 3–10) to NULL, and rejects frames from unknown or inactive nodes. DS18B20 fault codes (−127 and 85 °C) are passed through so that a database trigger can clean them. For sensor modules that report moisture on a 0–100 % dry-to-wet scale, the bridge can rescale the value to volumetric water content.

C. Relational Database

The Review-1 entity-relationship design had seven tables; the final schema has the nine 3NF tables of Table I [28]. It uses primary and foreign keys with ON DELETE rules, UNIQUE constraints and CHECK constraints for value ranges, phone format and status/date consistency [29].

TABLE I. Database Tables

Table	Stores
farmers	farmer identity and contact
plots	field area, soil type, location
crops	season, family, N demand, learned P5–P95 ranges
sensor_nodes	node, plot, fitted sensors, last seen
soil_health_logs	every reading with source and quality score
cultivation_cycles	sowing, harvest, status, yield
treatments	fertiliser/pesticide, N applied, cost
recommendation_runs	model used, window, VNBI, season
recommendation_items	ranked crops of each run

Eight views provide the latest reading and pH per plot, the 7-day conditions, the last crop, cycle performance, node health, a plot overview and v_recommended_crops, a rule-based recommender that joins each plot's conditions to the crop ranges. Three triggers enforce the rotation rule (an annual crop may not follow itself, and a plot may not have two growing cycles), null DS18B20 fault codes on insert, and update the node's last-seen time. Four stored procedures start a cycle, record a harvest inside a transaction with rollback, return a plot's history, and produce a yield report (total yield, actual versus catalogue maturation time and yield per acre). The function fn_virtual_n_index computes the VNBI of Section V. A generated demonstration dataset contains 10 farmers, 16 plots, 22 crops, 6 sensor nodes, 2,942 soil logs (10 days at one reading per 30 min per node), 62 cultivation cycles and 151 treatments.

D. Web Application

The Flask website guides a farmer through four steps: the field and its live readings, optional manual soil data, plain-language crop advice, and the crops grown and harvested. Choosing a recommended crop saves the run and starts a cultivation cycle through sp_start_cycle. Registering a farmer, field, sensor node and first reading happens in one transaction. Administrative pages and a JSON API expose the same data.

IV. Dataset and Models

A. Sensor-Oriented Dataset

The models are trained on the public Crop Recommendation dataset [30]: 2,200 rows, 22 crops, 100 rows per crop, with N, P, K, temperature, humidity, pH and rainfall. Because the node has no N-P-K sensor, N, P, K and rainfall were dropped, and three columns the node does measure were derived. Soil type is sampled per crop from Indian crop–soil associations based on NCERT geography [31]; this mapping is a team assumption. Volumetric moisture is

$$\theta = WP + (FC - WP)(1 - \exp(-R/R_0)) + \varepsilon, \qquad (1)$$

where FC and WP are the mid-points of field capacity and wilting point from FAO-56 Table 19 [32] for the texture of each soil type (e.g., alluvial 27/12 %, black 36/22 %), R is rainfall, $R_0 = 120$ mm and $\varepsilon \sim N(0, 1.2)$. Soil temperature is

$$T_{soil} = T_{air} + 1.5 - 0.08(\theta - 20) + \varepsilon, \qquad (2)$$

with $\varepsilon \sim N(0, 0.8)$. Values are rounded to the sensor resolution (DHT11 1 °C and 1 %RH, DS18B20 0.0625 °C). The features are moisture, soil temperature, air temperature, humidity, soil type and, optionally, pH. All experiments use a stratified 80/20 split with seed 42 (440 test rows) and stratified 5-fold cross-validation (CV).

Table II (Exp 1) measures the cost of having no N-P-K sensor with a Random Forest on the original dataset. Accuracy falls from 0.996 to 0.742 when N, P, K and rainfall are removed, so the sensor-only problem is substantially harder.

TABLE II. Effect of Removing N-P-K (Exp 1, Random Forest)

Features	CV accuracy
N, P, K, temperature, humidity, pH, rainfall	0.996
temperature, humidity, pH, rainfall	0.962
temperature, humidity, rainfall	0.920
temperature, humidity, pH	0.742

B. Model Selection

Nine classifiers from scikit-learn [33] were compared on both feature sets (Exp 2, Table III). The soft-voting ensemble of Random Forest, Histogram Gradient Boosting and an RBF SVM had the highest CV accuracy with pH, and the Random Forest was the strongest single model without pH.

TABLE III. CV Accuracy of the Compared Models (Exp 2)

Model	With pH	Without pH
Soft Voting (RF+HGB+SVM)	**0.895**	0.833
Hist Gradient Boosting	0.888	0.818
Random Forest	0.878	**0.831**
SVM (RBF)	0.872	0.819
Extra Trees	0.870	0.817
Decision Tree	0.820	0.772
K-Nearest Neighbours	0.811	0.795
Logistic Regression	0.732	0.671
Gaussian Naive Bayes	0.595	0.546

An extended comparison of 16 models (Exp 7, adding MLP, LDA, QDA, AdaBoost, Bagging, Gradient Boosting and Stacking) fixed the deployed models. With pH, Soft Voting had the highest CV accuracy and beat Histogram Gradient Boosting in 4 of 5 folds; Stacking reached 0.892. Without pH, Stacking (0.834) and Soft Voting (0.833) beat the Random Forest in only 2 of 5 folds, so the simpler Random Forest was kept. A nested-CV regularisation study (Exp 8) improved Logistic Regression by 0.064 with pH and 0.054 without pH (5 of 5 folds), and Histogram Gradient Boosting and SVM by 0.011 and 0.010 without pH (5 of 5 folds), but changed the Random Forest by at most 0.003 and lowered Histogram Gradient Boosting with pH by 0.005, so the deployment was not changed.

Both selected models were calibrated with isotonic regression [34] using 5-fold cross-validation. On the hold-out set the calibrated models reach 0.902 accuracy and 0.993 top-3 accuracy with pH, and 0.814 and 0.986 without pH; the website therefore always shows a ranked top three. Permutation importance on the hold-out set ranks humidity first (0.639 with pH), followed by moisture (0.247), pH (0.200), soil type (0.148), air temperature (0.113) and soil temperature (0.060). Soil temperature adds little because it was derived from air temperature by (2).

V. Proposed Methods

A. Quality-Weighted Temporal Consensus (QW-TC)

Instead of classifying the latest reading, the engine classifies a consensus feature vector over the last 7 days of readings x_i:

$$\bar{x} = \Sigma\, q_i\, w_i\, x_i \,/\, \Sigma\, q_i\, w_i. \qquad (3)$$

The quality score q_i starts at 1 and is set to 0 for a missing, out-of-range or fault-code value. A robust spike test compares each value with the window median using the median absolute deviation; a z-score above 3.5 multiplies q_i by $\max(0, 1 - (z - 3.5)/3.5)$. Moisture below 1 % (probe out of soil) multiplies q_i by 0.1, and a soil–air temperature difference above 15 °C by 0.5. The recency weight is day-blocked, $w_i = \exp(-24\, d_i/\tau)$ with d_i the whole days before the latest reading and $\tau = 168$ h, so all readings of one day share a weight and the day/night balance is kept. Quality scores are written back to soil_health_logs, and the 7-day view excludes rejected readings. If no reading in the window has a soil temperature (no probe fitted), it is estimated with (2) without noise.

B. Adaptive Model Selection

If the plot has a valid pH (between 3 and 10, at most 180 days old, from the probe or typed in), the with-pH model is used; otherwise the sensor-only model is used instead of imputing pH.

C. Calibration and Abstention

An Isolation Forest [35] (300 trees, contamination 0.01) trained on the training features marks inputs unlike any training condition as out-of-distribution. If the top calibrated probability is below 0.50, the run is marked low-confidence and the website advises a laboratory soil test.

D. Virtual Nitrogen Balance Index and Re-Ranking

Without an N-P-K sensor, soil nitrogen is estimated from the plot's last three cycles k = 0, 1, 2 (most recent first):

$$\mathrm{VNBI} = \Sigma\ 0.6^{k}\ (e(c_k) + \min(1, N_k/40)), \qquad (4)$$

where e(c) is +1 for a nitrogen-fixing legume, 0 for a low, −0.5 for a medium and −1 for a high nitrogen-demand crop, and N_k is the nitrogen applied in cycle k in kg per acre, taken from the treatments table. The same formula is implemented in Python and as fn_virtual_n_index with a window function. The final score is

$$S = p \times f_{season} \times f_{rotation} \times f_N, \qquad (5)$$

where p is the calibrated probability. $f_{season} = 0.6$ for an off-season annual crop. $f_{rotation}$ is 0 for the same annual crop as last cycle (also enforced by the trigger), 0.8 for the same family and 1.15 for a legume after a cereal or fibre crop. When VNBI < −0.75, f_N is 0.85 for high-N crops and 1.15 for legumes; when VNBI > 0.75, it is 1.1 for high-N crops. These factors are rule-based.

E. Explainable Fit Card

For each crop, the P5–P95 range of every measurement is learned from the training rows and stored in the crops table. The website shows whether each consensus value lies inside, below or above the range, and v_recommended_crops uses the same ranges as a transparent SQL baseline.

VI. Experiments and Results

A. Robustness to Sensor Faults (Exp 3)

Each test row was taken as a plot's daily-mean condition, and a 7-day window of 42 readings (one every 4 h) was simulated around it with diurnal cycles (air temperature ±4 °C, humidity ±8 %, soil temperature ±1.5 °C), slow moisture drift and datasheet-level noise. Each reading became faulty with probability f, chosen from DS18B20 disconnected (−127 °C), DS18B20 power-on reset (85 °C), probe out of soil (0 %), a humidity spike (±30 %) and a moisture spike (±15 %).

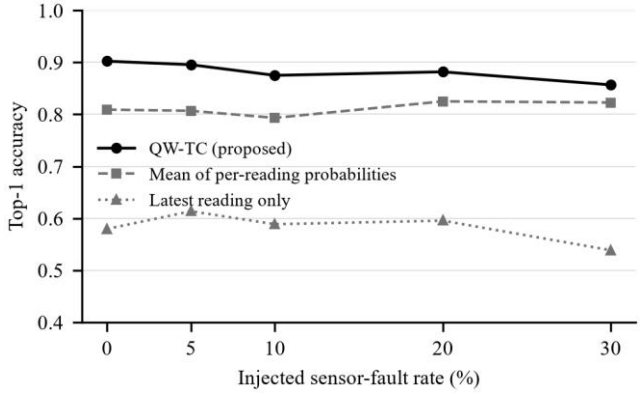


Fig. 2. Top-1 accuracy versus injected fault rate (Exp 3, with-pH model, 440 test rows).

Fig. 2 shows that classifying the latest reading gives 0.539–0.614, whereas QW-TC gives 0.857–0.902 for fault rates of 0–30 %. Averaging per-reading class probabilities gives only 0.793–0.825, because single readings carry the day/night swing while crop requirements describe daily-mean conditions. An hourly recency decay (τ = 72 h) gave 0.843–0.884, lower than day-blocked weighting at every fault level. With 440 test rows the standard error is about ±0.017, so differences below about 3 points between the consensus variants are within noise; the large gain is from snapshot to consensus.

B. Missing pH (Exp 4)

TABLE IV. Strategies When pH Is Missing (Exp 4)

Strategy	Accuracy
With-pH model, pH = 7.0	0.677
With-pH model, pH = training median	0.791
Sensor-only model (proposed)	**0.814**
Reference: pH measured	0.902

Table IV shows that switching to the sensor-only model is more accurate than either imputation.

C. Calibration and Abstention (Exp 2 and Exp 5)

TABLE V. Effect of Isotonic Calibration (Hold-Out Set)

Model	ECE before	ECE after	Accuracy after
Soft Voting (with pH)	0.086	0.022	0.902
Random Forest (no pH)	0.074	0.051	0.814

Calibration reduced the expected calibration error (ECE, 15 bins) for both models (Table V). Fig. 3 shows the risk–coverage curve: at the chosen threshold of 0.50, 97.0 % of recommendations are answered with 91.8 % accuracy, and the abstained 3.0 % are only 38.5 % accurate. The Isolation Forest flagged 0.45 % of real test readings but 70.1 % of 1,000 random in-range readings.

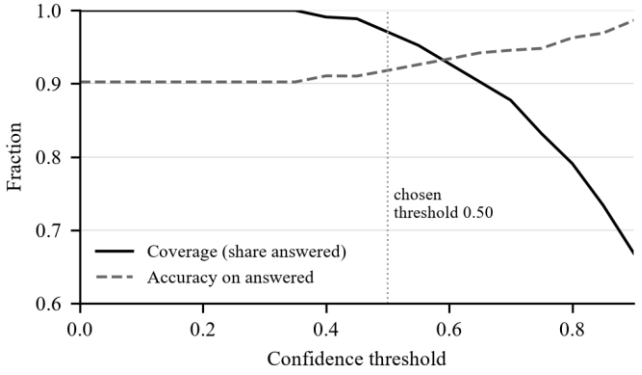


Fig. 3. Coverage and accuracy versus confidence threshold (Exp 5, with-pH model).

D. Context-Aware Re-Ranking (Exp 6)

One real test reading (row 150: moisture 17.4 %, soil temperature 23.1 °C, air temperature 21 °C, humidity 65 %, pH 6.67, alluvial soil) was ranked under five plot histories. The model alone ranks maize (0.62) above lentil (0.38). Table VI gives the context-aware top two.

TABLE VI. RE-RANKING OF ONE READING UNDER FIVE HISTORIES (EXP 6)

Scenario	VNBI	Top two (score)
No history, July	–	maize 0.62, lentil 0.23
Last crop maize, July	−1.00	lentil 0.30, apple 0.00
Rice, maize, cotton, no fertiliser, July	−1.96	maize 0.52, lentil 0.30
Same, 50 kg N/acre each cycle, July	0.00	maize 0.62, lentil 0.26
No history, November	–	lentil 0.38, maize 0.37

Repeating maize is blocked, a depleted nitrogen history lowers maize from 0.62 to 0.52, recorded urea removes that penalty, and in November (Rabi) lentil overtakes the Kharif crop maize. This is a case study; whether the re-ranking improves yield needs multi-season field data.

E. Prototype Test

The data path was tested on 13 October 2026 with an Arduino Uno carrying a soil-moisture sensor and a DHT-type air sensor; the DS18B20 was not yet fitted. The module reported moisture on a 0–100 % dry-to-wet scale, so the bridge rescaled it to volumetric content (100 % → 45 %). Readings arrived about every 10 s: 0 % with the probe in air and 96–100 % (stored as 43–45 %) in freshly watered soil. When the probe was pulled out, the 0 % readings that followed received zero weight in QW-TC, so the consensus moisture stayed at 44.2 % and the advice did not change. With pH 5.6 entered by hand and 55 valid readings, the with-pH model ranked pigeon peas first (probability 0.611, final score 0.367 after the off-season factor 0.6 in October) and flagged the moisture as above the range of every top crop, as expected for saturated soil. This test shows the working data path, not field accuracy.

VII. LIMITATIONS

Moisture, soil temperature and soil type in the training data are derived from a public dataset with documented formulas, not measured in the field, and the collection method of that dataset is not documented in detail by its author. The Exp 3 sensor windows are simulated around dataset rows. The VNBI weights and re-ranking factors are heuristic and were evaluated by case study, not learned from yield data. The agronomic constants (seasons, maturation days, typical yields) still need to be checked against ICAR and Government of India sources, and the prototype test used one node without a soil-temperature probe.

VIII. CONCLUSION AND FUTURE WORK

AgriSmart shows that a relational database fed by a low-cost node without an N-P-K sensor can support reliable and explainable crop advice when the decision uses the plot's stored history. QW-TC raised top-1 accuracy under simulated faults from 0.539–0.614 to 0.857–0.902, the adaptive model switch beat pH imputation (0.814 versus 0.677–0.791), and calibration with abstention made the reported confidence meaningful. Agronomic rules such as crop rotation are enforced inside the database by a trigger, and nitrogen status is estimated in SQL from recorded history. Future work is to fit the soil-temperature probe, collect real field readings and multi-season yields to validate the models and the VNBI, and verify the agronomic constants against official sources.

REFERENCES

[1] O. Elijah, T. A. Rahman, I. Orikumhi, C. Y. Leow, and M. N. Hindia, "An overview of Internet of Things (IoT) and data analytics in agriculture: Benefits and challenges," IEEE Internet of Things Journal, vol. 5, no. 5, pp. 3758–3773, Oct. 2018.

[2] S. Wolfert, L. Ge, C. Verdouw, and M.-J. Bogaardt, "Big data in smart farming – A review," Agricultural Systems, vol. 153, pp. 69–80, 2017.

[3] S. Pudumalar, E. Ramanujam, R. H. Rajashree, C. Kavya, T. Kiruthika, and J. Nisha, "Crop recommendation system for precision agriculture," in Proc. IEEE 8th Int. Conf. Advanced Computing (ICoAC), Chennai, India, 2016, pp. 32–36.

[4] Maxim Integrated, DS18B20 Programmable Resolution 1-Wire Digital Thermometer, datasheet.

[5] M. Ayaz, M. Ammad-Uddin, Z. Sharif, A. Mansour, and E.-H. M. Aggoune, "Internet-of-Things (IoT)-based smart agriculture: Toward making the fields talk," IEEE Access, vol. 7, pp. 129551–129583, 2019.

[6] C. Tzounis, N. Katsoulas, T. Bartzanas, and C. Kittas, "Internet of Things in agriculture, recent advances and future challenges," Biosystems Engineering, vol. 164, pp. 31–48, 2017.

[7] A. Khanna and S. Kaur, "Evolution of Internet of Things (IoT) and its significant impact in the field of precision agriculture," Computers and Electronics in Agriculture, vol. 157, pp. 218–231, 2019.

[8] M. S. Farooq, S. Riaz, A. Abid, K. Abid, and M. A. Naeem, "A survey on the role of IoT in agriculture for the implementation of smart farming," IEEE Access, vol. 7, pp. 156237–156271, 2019.

[9] G. Codeluppi, A. Cilfone, L. Davoli, and G. Ferrari, "LoRaFarM: A LoRaWAN-based smart farming modular IoT architecture," Sensors, vol. 20, no. 7, p. 2028, 2020.

[10] A. Villa-Henriksen, G. T. C. Edwards, L. A. Pesonen, O. Green, and C. A. G. Sørensen, "Internet of Things in arable farming: Implementation, applications, challenges and potential," Biosystems Engineering, vol. 191, pp. 60–84, 2020.

[11] A. Kamilaris, A. Kartakoullis, and F. X. Prenafeta-Boldú, "A review on the practice of big data analysis in agriculture," Computers and Electronics in Agriculture, vol. 143, pp. 23–37, 2017.

[12] S. A. Bhat and N.-F. Huang, "Big data and AI revolution in precision agriculture: Survey and challenges," IEEE Access, vol. 9, pp. 110209–110222, 2021.

[13] R. Sharma, S. S. Kamble, A. Gunasekaran, V. Kumar, and A. Kumar, "A systematic literature review on machine learning applications for sustainable agriculture supply chain performance," Computers & Operations Research, vol. 119, p. 104926, 2020.

[14] Q. Zhang, M. Wang, and N. Wang, "Precision agriculture – A worldwide overview," Computers and Electronics in Agriculture, vol. 36, no. 2–3, pp. 113–132, 2002.

[15] M. Dhanaraju, P. Chenniappan, K. Ramalingam, S. Pazhanivelan, and R. Kaliaperumal, "Smart farming: Internet of Things (IoT)-based sustainable agriculture," Agriculture, vol. 12, no. 10, p. 1745, 2022.

[16] K. G. Liakos, P. Busato, D. Moshou, S. Pearson, and D. Bochtis, "Machine learning in agriculture: A review," Sensors, vol. 18, no. 8, p. 2674, 2018.

[17] A. Kamilaris and F. X. Prenafeta-Boldú, "Deep learning in agriculture: A survey," Computers and Electronics in Agriculture, vol. 147, pp. 70–90, 2018.

[18] T. van Klompenburg, A. Kassahun, and C. Catal, "Crop yield prediction using machine learning: A systematic literature review," Computers and Electronics in Agriculture, vol. 177, p. 105709, 2020.

[19] J. H. Jeong et al., "Random forests for global and regional crop yield predictions," PLoS ONE, vol. 11, no. 6, p. e0156571, 2016.

[20] P. S. Maya Gopal and R. Bhargavi, "A novel approach for efficient crop yield prediction," Computers and Electronics in Agriculture, vol. 165, p. 104968, 2019.

[21] D. Elavarasan and P. M. D. Vincent, "Crop yield prediction using deep reinforcement learning model for sustainable agrarian applications," IEEE Access, vol. 8, pp. 86886–86901, 2020.

[22] C. Kamienski et al., "Smart water management platform: IoT-based precision irrigation for agriculture," Sensors, vol. 19, no. 2, p. 276, 2019.

[23] S. R. Nandurkar, V. R. Thool, and R. C. Thool, "Design and development of precision agriculture system using wireless sensor network," in Proc. IEEE Int. Conf. Automation, Control, Energy and Systems (ICAETR), 2014, pp. 1–6.

[24] Y. Kim, R. G. Evans, and W. M. Iversen, "Remote sensing and control of an irrigation system using a distributed wireless sensor network,"

IEEE Transactions on Instrumentation and Measurement, vol. 57, no. 7, pp. 1379–1387, Jul. 2008.

[25] J. Muangprathub, N. Boonnam, S. Kajornkasirat, N. Lekbangpong, A. Wanichsombat, and P. Nillaor, "IoT and agriculture data analysis for smart farm," Computers and Electronics in Agriculture, vol. 156, pp. 467–474, 2019.

[26] S. Fenz, "AI- and data-driven crop rotation planning," Computers and Electronics in Agriculture, vol. 212, p. 108160, 2023.

[27] Aosong Electronics, DHT11 Humidity & Temperature Sensor, datasheet.

[28] R. Elmasri and S. B. Navathe, Fundamentals of Database Systems, 7th ed. Hoboken, NJ, USA: Pearson, 2016.

[29] Oracle Corporation, "MySQL 8.0 Reference Manual: Stored Programs and Views," MySQL Documentation, 2024. [Online]. Available: https://dev.mysql.com/doc/refman/8.0/en/

[30] A. Ingle, "Crop Recommendation Dataset," Kaggle, 2020. [Online]. Available: https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset

[31] NCERT, Contemporary India II: Textbook in Geography for Class X, Ch. 1 "Resources and Development" and Ch. 4 "Agriculture." New Delhi, India: NCERT.

[32] R. G. Allen, L. S. Pereira, D. Raes, and M. Smith, Crop Evapotranspiration: Guidelines for Computing Crop Water Requirements, FAO Irrigation and Drainage Paper 56. Rome, Italy: FAO, 1998.

[33] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[34] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," in Proc. 8th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, 2002, pp. 694–699.

[35] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in Proc. 8th IEEE Int. Conf. Data Mining (ICDM), 2008, pp. 413–422.